in Tamil Nadu State was taken and their average weight was found to be 120 lbs with a standard deviation of 30 lbs. Calculate standard error of mean.

19. A random sample of 60 observations was drawn from a large population and its standard deviation was found to be 2.5. Calculate the suitable standard error that this sample is taken from a population with standard deviation 3?

20. In a sample of 400 population from a village 230 are found to be eaters of vegetarian items and the rest non-vegetarian items. Compute the standard error assuming that both vegetarian and non-vegetarian foods are equally popular in that village?

Statistical Inference

One of the main objectives of any statistical investigation is to draw inferences about a population from the analysis of samples drawn from that population. Statistical Inference provides us how to estimate a value from the sample and test that value for the population. This is done by the two important classifications in statistical inference,

(i) Estimation;

(ii) Testing of Hypothesis

8.2 Estimation:

It is possible to draw valid conclusion about the population parameters from sampling distribution. Estimation helps in estimating an unknown population parameter such as population mean, standard deviation, etc., on the basis of suitable statistic computed from the samples drawn from population.

Estimation:

Definition 8.4

The method of obtaining the most likely value of the population parameter using statistic is called estimation.

Estimator:

Definition 8.5

Any sample statistic which is used to estimate an unknown population parameter is called an estimator ie., an estimator is a sample statistic used to estimate a population parameter.

Estimate:

Definition 8.6

When we observe a specific numerical value of our estimator, we call that value is an estimate. In other words, an estimate is a specific observed value of a statistic.

Characteristic of a good estimator

A good estimator must possess the following characteristic:

(i) Unbiasedness (ii) Consistency (iii) Efficiency (iv) Sufficiency.

(i) Unbiasedness: An estimator $T_n = T(x_1, x_2, ..., x_n)$ is said to be an unbiased estimator of $\gamma(\theta)$ *if* $E(T_n) = \gamma(\theta)$, for all $\theta \varepsilon \Theta$ (parameter space), (i.e) An estimator is said to be unbiased if its expected value is equal to the population parameter. Example: $E(\bar{x}) = \varphi$

(ii) Consistency: An estimator $T_n = T(x_1, x_2, ..., x_n)$ is said to be consistent estimator of $\gamma(\theta)$, if T_n converges to $\gamma(\theta)$ in Probability, i.e., $T_n \xrightarrow{P} \gamma(\theta) as\, n \to \infty$, for all $\theta\ \varepsilon \Theta$.

(iii) Efficiency: If T_1 is the most efficient estimator with variance V_1 and T_2 is any other estimator with variance V_2, then the efficiency E of T_2 is defined as $E = \frac{V_1}{V_2}$ Obviously, E cannot exceed unity.

(iv) Sufficiency: If $T = t(x_1, x_2, ..., x_n)$ is an estimator of a parameter θ, based on a sample $x_1, x_2, ..., x_n$ of size n from the

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